

NUIQSUT, AK, USA

WMO: 703644

Lat: 70.212N

Lon: 151.002W

Elev: 17

StdP: 101.12

Time Zone: -9.00 (NAK)

Period: 99-19

WBAN: 27515

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -42.4	(c) -40.3	(d) -37.4	(e) 0.1	(f) -32.9	(g) -36.9	(h) 0.1	(i) -33.1	(j) 16.4	(k) -19.7	(l) 15.0	(m) -18.9	(n) 2.6	(o) 200	(p) 1.014

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.7	21.0	14.7	18.2	13.1	16.1	12.2	15.0	19.6	13.7	17.6	12.4	15.7	4.5	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.7	9.2	16.6	11.4	8.4	15.4	10.2	7.7	14.3	41.8	19.7	38.2	17.4	35.0	15.9	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.8	11.9	10.6	-43.4	24.8	3.1	2.2	-45.7	26.5	-47.5	27.8	-49.2	29.0	-51.4	30.6
			-36.1	17.2	0.7	2.6	-36.6	19.1	-37.0	20.6	-37.4	22.0	-37.8	23.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-9.8	-25.8	-24.7	-25.1	-16.1	-4.4	5.7	9.8	6.8	2.0	-6.3	-16.7	-23.1		
	DBStd	14.51	7.94	8.80	7.50	6.73	5.43	4.73	3.88	3.95	3.62	4.73	7.02	7.84		
	HDD10.0	7294	1111	972	1089	783	448	148	54	118	241	504	800	1026		
	HDD18.3	10252	1369	1206	1348	1033	706	380	266	357	490	763	1050	1284		
	CDD10.0	85	0	0	0	0	0	18	46	19	1	0	0	0		
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0		
	CDH23.3	13	0	0	0	0	0	3	8	1	0	0	0	0		
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0		
(14) Wind	WSAvg	4.7	4.8	4.9	4.8	4.7	4.8	5.0	4.6	4.3	4.4	4.7	4.6	4.5		
(15) Precipitation	PrecAvg	205	11	10	7	6	9	14	30	38	28	25	14	12		
	PrecMax	347	34	28	42	16	21	37	60	93	92	52	39	36		
	PrecMin	116	2	1	0	0	1	3	6	11	5	11	4	2		
	PrecStd	45	7	6	8	4	6	9	12	19	20	10	8	8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.9	-1.6	-3.2	2.5	9.8	22.9	24.2	22.2	16.4	4.5	0.0	-2.2		
		MCWB	-1.4	-2.2	-3.9	1.4	6.3	14.2	15.8	15.9	10.9	3.4	-0.5	-3.4		
	2%	DB	-6.0	-5.2	-7.2	-1.1	6.1	19.2	22.0	18.6	12.7	1.9	-3.0	-5.9		
		MCWB	-6.5	-5.9	-7.4	-1.7	4.1	12.8	15.1	13.7	9.3	1.2	-3.7	-6.8		
	5%	DB	-12.2	-8.2	-11.3	-3.7	3.7	16.4	19.2	16.2	9.7	0.5	-5.2	-9.2		
		MCWB	-12.7	-8.6	-11.7	-4.3	2.5	11.7	13.9	12.4	7.8	0.1	-5.7	-9.6		
	10%	DB	-15.8	-12.3	-15.4	-6.3	2.1	13.7	17.1	13.5	7.0	-0.7	-7.0	-12.8		
		MCWB	-16.0	-12.8	-15.7	-6.7	1.3	10.2	13.0	11.2	5.9	-1.1	-7.4	-13.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.2	-2.1	-3.3	1.5	6.5	15.1	17.4	16.3	11.3	3.5	-0.5	-3.1		
		MCDB	-0.7	-1.6	-2.9	2.4	9.1	21.6	23.3	20.8	15.2	4.5	-0.1	-2.4		
	2%	WB	-5.6	-5.2	-7.2	-1.7	4.1	13.2	15.6	14.2	9.6	1.3	-3.6	-6.2		
		MCDB	-5.1	-4.6	-6.8	-1.3	5.8	18.4	20.5	17.6	12.8	1.7	-3.2	-5.5		
	5%	WB	-11.6	-8.2	-11.1	-4.1	2.5	11.9	14.3	12.7	7.9	0.1	-5.7	-9.4		
		MCDB	-11.2	-7.8	-10.7	-3.6	3.6	16.1	18.5	15.4	9.8	0.4	-5.3	-9.0		
	10%	WB	-15.1	-11.6	-14.7	-6.7	1.3	10.4	13.1	11.3	5.9	-1.0	-7.3	-12.9		
		MCDB	-14.7	-11.0	-14.1	-6.4	2.0	13.8	16.6	13.6	7.0	-0.7	-6.9	-12.5		
(35) Mean Daily Temperature Range	MDBR		7.3	7.6	7.2	7.8	5.2	7.9	8.7	6.7	5.2	5.3	7.3	7.0		
	5% DB	MCDBR	11.9	11.2	9.1	8.3	7.4	12.6	13.1	11.6	10.9	5.0	7.9	9.7		
		MCWBR	11.5	10.8	8.9	7.9	5.6	7.3	7.2	7.0	7.4	4.6	7.5	9.4		
	5% WB	MCDBR	12.1	11.3	9.3	8.2	7.1	12.5	12.1	10.6	10.5	4.8	7.7	9.6		
		MCWBR	11.9	11.0	9.1	7.8	5.5	7.6	7.3	6.7	7.2	4.4	7.5	9.4		
(40) Clear-Sky Solar Irradiance	taub		N/A	0.202	0.239	0.297	0.326	0.331	0.343	0.335	0.290	0.216	N/A	N/A		
	taud		N/A	2.064	2.236	2.146	2.200	2.404	2.449	2.497	2.536	2.159	N/A	N/A		
	Ebn at Noon		0	543	790	829	848	861	835	798	743	508	0	0		
	Edh at Noon		0	37	67	104	113	96	89	76	54	32	0	0		
(44) All-Sky Solar Radiation	RadAvg		0.00	0.36	1.96	4.32	5.93	6.18	5.09	2.98	1.50	0.45	0.00	0.00		
	RadStd		0.00	0.03	0.07	0.19	0.32	0.42	0.40	0.30	0.14	0.06	0.00	0.00		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	+1.14	N/A	N/A	N/A	N/A	N/A	-501	-494	N/A	N/A

Nomenclature: See separate page